Zhiji Han

Lecturer in Control Science and Engineering

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Short bio: Zhiji Han received his B.Eng. degree in automation and Ph.D. degree in control science and engineering from the University of Science and Technology Beijing (USTB), Beijing, China, in 2019 and 2024, respectively. Since August 2024, he has been a Lecturer and Postdoctoral Researcher with the College of Engineering, Ocean University of China (OUC), Qingdao, China. His research interests include modeling and control of distributed parameter systems (infinite dimensional systems); dynamical modeling and analysis of flexible structures; vibration stabilization, shape regulation, and trajectory tracking of flexible structures. He has co-authored 28 SCI journal publications (among which 10 in IEEE Transactions, 1 in Automatica), with 775 Google Scholar citations. His publications include 2 ESI Highly Cited Papers and 1 ESI Hot Paper. He holds 8 granted/pending invention patents. He serves as an Early Career Editorial Board Member for *Biomimetic Intelligence and Robotics*, and has been an invited session chair at several international conferences including IFAC HMS 2025, YAC 2025–2026, and CAC 2024.

Current Positions

Since 2024 **Lecturer** in Control Science and Engineering at the College of Engineering, Ocean University of China (OUC), Qingdao, China.

Since 2024 **Postdoctoral Researcher** at the College of Engineering, Ocean University of China (OUC), Qingdao, China.

Education

2019–2024 **Doctoral degree** in Control Science and Engineering from the University of Science and Technology Beijing (USTB), Beijing, China.
Thesis: *Partial Differential Equation Based Modeling and Control of Rope-Driven Soft Arm.*

2015–2019 **Bachelor of Engineering** in Automation from the University of Science and Technology Beijing (USTB), Beijing, China.

Research Interests

Keywords: Distributed parameter systems (infinite dimensional systems); PDE-based modeling and control; flexible structures; continuum systems; vibration control; adaptive control.

Author of 28 SCI journal publications and 8 conference papers on modeling and control of distributed parameter systems and flexible structures, with applications to soft robots, aerial refueling hoses, wind turbine towers, marine risers, high-rise buildings, and spacecraft.

Research Projects as Principal Investigator

2026–2028 **National Natural Science Foundation of China (NSFC)** — Category C.
Distributed Adaptive Safety Control of Flexible Arm System Based on Cosserat Rod.
Grant No. 62503443.

2025–2026 **National Foreign Expert Program** — Category H.
Modeling and Control Research of Flexible Robotic Arm Systems for Disturbance Compensation and Performance Constraints.
Grant No. 110000206320258005.

2025–2026 **China Postdoctoral Science Foundation (CPSF)**.
Modeling and Coordinated Control of Mobile Continuum Grasping Systems under Multi-Constraints.
Grant No. 2025M771694.

2025–2026 **Postdoctoral Fellowship Program of CPSF** — Category C.
Modeling and Control of an Underwater Cable-Driven Flexible Manipulator with Non-Collocated Actuators and Sensors.
Grant No. GZC20251171.

2025–2027 **Postdoctoral Innovation Program of Shandong Province** — First-Class Funding.
Modeling and Coordinated Control of Parallel Continuum Arm Systems.
Grant No. SDCX-ZG-202501023.

2025–2027 **Qingdao Natural Science Foundation**.
Cooperative Handling Control of Underwater Dual Flexible Arm Systems.
Grant No. 25-1-1-116-zyyd-jch.

2024–2026 **Qingdao Postdoctoral Program**.
Dynamic Modeling and Shape Control of a Multi-Arm Flexible Mining System for Uneven Seabed Environments.
Grant No. QDBSH20250102058.

2025–2026 **Fundamental Research Funds for the Central Universities**.
Partial Differential Equation Based Modeling and Control of a Class of Soft Robots.
Grant No. 202513025.

Awards and Honors

2024 **8th China Association for Science and Technology (CAST) Outstanding Sci-Tech Paper Award.**

2023 **Hot Paper Award** — *SCIENCE CHINA Information Sciences.*

2023 **Outstanding Graduate Student** — Beijing Association of Automation.

2022–2025 **Excellence in Review** — *IEEE Journal of Ocean Engineering.*

2023 **Outstanding Reviewer** — *SCIENCE CHINA Information Sciences.*

Editorial Activities

Since 2026 **Early Career Editorial Board Member** — *Biomimetic Intelligence and Robotics.*

Conference Organization — Invited Session Chair

2026 **Co-Chair**, Invited Session 52: "Modeling, Control and Optimization of Spatio-Temporal Big-Data Intelligent Systems," 41st Youth Academic Annual Conference of Chinese Association of Automation (YAC 2026), Changsha, China, May 8–10.

2025 **Co-Chair**, Special Session: "Modeling and Control Theory of Complex Systems," 16th IFAC Symposium on Analysis, Design and Evaluation of Human-Machine Systems (IFAC HMS 2025), Beijing, China, November 18–21.

2025 **Co-Chair**, Invited Session 16: "Intelligent Control of Unmanned Systems in Multi-Constraints Environment," 40th Youth Academic Annual Conference of Chinese Association of Automation (YAC 2025), Zhengzhou, China, May 17–19.

2024 **Co-Chair**, Oral Presentation Session: "Learning, Optimization, and Decision-Making Based on Big Data," 2024 China Automation Congress (CAC 2024), Qingdao, China, November 1–3.

Reviewer Activities

Reviewer for international journals (180 verified reviews), including: *IEEE Transactions on Automatic Control*, *Automatica*, *IEEE Transactions on Automation Science and Engineering*, *IEEE Transactions on Cybernetics*, *IEEE Transactions on Industrial Electronics*, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *IEEE Transactions on Vehicular Technology*, *IEEE/ASME Transactions on Mechatronics*, *IEEE/CAA Journal of Automatica Sinica*, *IEEE Journal of Ocean Engineering*, *SCIENCE CHINA Information Sciences*, among others.

Patents

8 invention patents applied for / authorized in the field of modeling and control of flexible structures and soft robots.

Scientific Production

28	SCI Journal Papers	**8**	International Conference Papers
8	Invention Patents	**775**	Google Scholar Citations

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Journal Distribution

2	IEEE Trans. on Automatic Control	**2**	IEEE Trans. on Systems, Man, & Cybernetics: Systems
2	IEEE Trans. on Fuzzy Systems	**1**	IEEE Trans. on Automation Science & Engineering
1	IEEE Trans. on Control Systems Technology	**1**	IEEE/ASME Trans. on Mechatronics
1	IEEE Trans. on Aerospace & Electronic Systems	**1**	IEEE/CAA Journal of Automatica Sinica
1	Automatica	**1**	Fuzzy Sets and Systems
1	Science China Information Sciences	**1**	Commun. Nonlinear Science & Numerical Simulation
1	Applied Mathematical Modelling	**1**	Journal of Field Robotics
1	Int. J. Robust & Nonlinear Control	**1**	Journal of Intelligent & Robotic Systems
1	Journal of the Franklin Institute	**3**	Asian Journal of Control
1	Biomimetics	**1**	Robotic Intelligence and Automation

Selected Publications

1. Zhiji Han, Zhijie Liu, Wen Kang, and Wei He, "Boundary feedback control of a nonhomogeneous wind turbine tower with exogenous disturbances," *IEEE Transactions on Automatic Control*, vol. 67, no. 4, pp. 1952–1959, 2022.
   (ESI Highly Cited Paper)
2. Zhiji Han, Zhijie Liu, Jun-Wei Wang, and Wei He, "Fault-tolerant control for flexible structures with partial output constraint," *IEEE Transactions on Automatic Control*, vol. 69, no. 4, pp. 2668–2675, 2024.
3. Zhiji Han, Zhijie Liu, Thomas Meurer, and Wei He, "PDE-based control synthesis for a planar cable-driven continuum arm," *Automatica*, vol. 163, p. 111600:1–111600:10, 2024.
4. Zhiji Han, Zhijie Liu, Wei He, and Guang Li, "Distributed parameter modeling and boundary control of an octopus tentacle-inspired soft robot," *IEEE Transactions on Control Systems Technology*, vol. 30, no. 3, pp. 1244–1256, 2022.
5. Zhiji Han, Zhijie Liu, Linghuan Kong, Liang Ding, Jun-Wei Wang, and Wei He, "Adaptive fuzzy control for a hybrid spacecraft system with spatial motion and communication constraints," *IEEE Transactions on Fuzzy Systems*, vol. 30, no. 8, pp. 3247–3256, 2022.
6. Zhiji Han, Zhijie Liu, Hongdu Wang, Yong Ren, Yidao Ji, and Wei He, "Control design for nonuniform continuum arm system using static force and in-domain velocity feedback," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 56, no. 3, pp. 1885–1894, 2026.
7. Wen Kang, Zhiji Han, Zhijie Liu, and Bao-Zhu Guo, "Fuzzy observer for 2-D parabolic equation with output time delay," *IEEE Transactions on Fuzzy Systems*, vol. 29, no. 11, pp. 3552–3560, 2021.
8. Zhijie Liu, Zhiji Han, and Wei He, "Adaptive fault-tolerant boundary control of an autonomous aerial refueling hose system with prescribed constraints," *IEEE Transactions on Automation Science and Engineering*, vol. 19, no. 4, pp. 2678–2688, 2022.
9. Zhijie Liu, Zhiji Han, Zhijia Zhao, and Wei He, "Modeling and adaptive control for a spatial flexible spacecraft with unknown actuator failures," *Science China Information Sciences*, vol. 64, p. 152208:1–152208:16, 2021.
   (ESI Highly Cited Paper; ESI Hot Paper)

Selected Conference Papers

1. Yuhua Song, Zhiji Han, Xiuyu He, and Wei He, "Modeling and boundary control design for a high-rise building structure," *58th IEEE Conference on Decision and Control (CDC)*, Nice, France, Dec. 11–13, 2019, pp. 2970–2975.
2. Huiyang Song, Zhiji Han, Zhijie Liu, Guang Li, and Wei He, "Non-collocated observer based control design for PDE-based multi-agent deployment," *13th Asian Control Conference (ASCC)*, Jeju, Korea, May 4–7, 2022, pp. 2061–2066.
3. Zhiji Han, Zhijie Liu, Thomas Meurer, and Wei He, "Robust fault-tolerant control of wave equation without estimations of plant and failure parameters," *4th IFAC Workshop on Control of Systems Governed by Partial Differential Equations (CPDE 2022)*, Kiel, Germany, Sep. 5–7, 2022, pp. 143–148.

4. Jiangtao Han, Zhiji Han, and Zhijie Liu, "Adaptive control for a constrained soft manipulator with prescribed performance," *3rd IFAC Workshop on Cyber-Physical & Human Systems (CPHS 2020)*, Beijing, China, Dec. 3–5, 2020, pp. 524–529.